

DFFR: DETR With Foreground-Guided Feature Refinement Network for End-to-End Underwater Object Detection

Gaoli Zhao , Kefei Zhang , *Student Member, IEEE*, Yuheng Wu , Zheng Liang , Wenyi Zhao , and Weidong Zhang , *Senior Member, IEEE*

Abstract—Underwater object detection is a vital technology for marine resource exploration, autonomous underwater vehicles, and environmental monitoring. Compared to land-based scenarios, this task is often hindered by challenging conditions, such as low contrast, object occlusion, and image blurring, which severely degrade detection performance. To effectively address these issues, we propose the transformer-based detectors with foreground-guided feature refinement network (DFFR). We first employ the lightweight high performance GPU network V2 (HGNetV2) backbone to extract multiscale deep semantic features, while the subsequent cross-modal encoder further enriches these representations by integrating both visual and textual information. We then introduce the foreground-guided feature refinement module, which enables interlayer feature fusion using high-level contextual guidance to significantly improve foreground recognition. Concurrently, we propose the location perception fusion strategy, which integrates multiscale features to enhance their consistency and discriminative capabilities. Finally, the refined features are passed through a fine-grained distribution refinement module for iterative optimization of the output distribution. Comprehensive experimental results demonstrate that DFFR outperforms existing state-of-the-art methods in multiple metrics, including average

precision (AP), AP₅₀, and AP₇₅, with improvements in AP of 2.1%, 2.9%, respectively on two underwater benchmark datasets, and AP₅₀ of improvement of 0.3% on the COCO dataset.

Index Terms—Cross-modality encoder, foreground-guided feature refinement (FFR), location perception fusion (LPF), underwater object detection (UOD).

I. INTRODUCTION

A. Background

WITH the rapid advancement of intelligent ocean technologies, underwater object detection (UOD) has become an indispensable component of applications, such as underwater inspection, marine ecological monitoring, and navigation for autonomous underwater vehicles (AUVs) [1], [2], [3]. However, compared to land-based scenes, underwater detection faces significant challenges from complex environmental factors, as illustrated in Fig. 1. Currently, two primary paradigms dominate object detection. Classical real-time frameworks, notably those from the You Only Look Once (YOLO) family [4], [5], [6], [7], are widely recognized for their high efficiency and robust community support. Despite relying on manually designed components, such as anchor box generation and nonmaximum suppression (NMS), these detectors continue to deliver leading performance. Meanwhile, end-to-end transformer-based detectors (DETRs) [8], [9], [10], [11] are becoming increasingly prominent, benefiting from their powerful global modeling capabilities and fully end-to-end training paradigm.

B. Motivation

Although the aforementioned models have achieved remarkable success in land-based industrial vision tasks, the complexity and variability inherent in underwater environments severely constrain their performance [12], [13]. Numerous enhancement strategies have been proposed to address these limitations, such as integrating image enhancement with feature alignment, incorporating attention mechanisms, and exploiting multiscale information to better handle object-scale variations [14]. Nevertheless, when applied to complex underwater environments, these methods still exhibit notable limitations, which hinder stable and precise detection results.

Received 8 October 2025; revised 14 November 2025 and 13 December 2025; accepted 26 December 2025. Date of publication 22 January 2026; date of current version 6 April 2026. This work was supported in part by the Training Program for Young Backbone Teachers in Higher Education Institutions of Henan Province under Grant 2025GGJS099, in part by the National Natural Science Foundation of China under Grant 62403066, in part by the China Postdoctoral Science Foundation under Grant 2024M750747, in part by the Key Specialized Research and Development Program of Science and Technology of Henan Province under Grant 242102211096 and Grant 242102210075, and in part by the Henan Provincial Science and Technology Research and Development Joint Foundation Project under Grant 235200810066. Paper no. TII-25-6691. (Corresponding author: Weidong Zhang.)

Gaoli Zhao, Kefei Zhang, and Yuheng Wu are with the School of Computer and Technology, Henan Institute of Science and Technology, Xinxiang 453003, China (e-mail: zhaogaoli@hist.edu.cn; zhangkefei@stu.hist.edu.cn; wyh@stu.hist.edu.cn).

Zheng Liang is with the School of Internet, Anhui University, Hefei 230093, China (e-mail: zliang@ahu.edu.cn).

Wenyi Zhao is with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing 102206, China (e-mail: 2013211876@bupt.edu.cn).

Weidong Zhang is with the School of Information Engineering, Henan Institute of Science and Technology, Xinxiang 453003, China, and also with the School of Electrical and Information Engineering, Zhengzhou University, Zhengzhou 450001, China (e-mail: zwd@hist.edu.cn).

Digital Object Identifier 10.1109/TII.2025.3649916

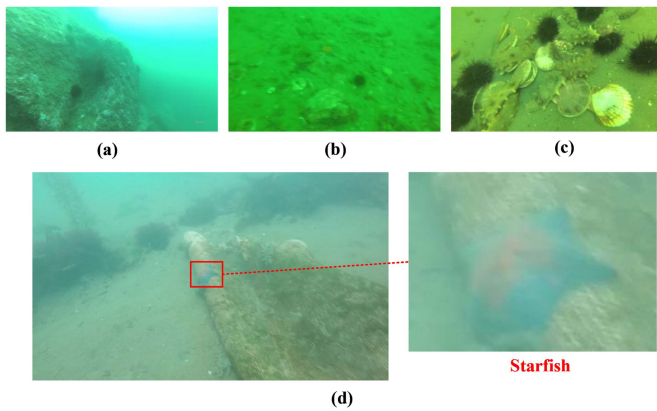


Fig. 1. Challenges in the underwater environment. (a) Complex underwater terrains create unbalanced light condition. (b) Low-contrast images blur the boundaries between scallops and holothurians. (c) Aquatic organisms often live in groups, which leads to occlusion. (d) Starfish has a color similar to its surroundings, making it difficult to detect. (a) Unbalanced light condition. (b) Low contrast. (c) Severe occlusion. (d) Camouflage and mimicry.

C. Analysis and Inference

What challenges fundamentally limit performance? To answer this question, we conducted an in-depth analysis of the feature representation and discrimination capabilities of existing detectors, as these aspects determine model robustness and reliability [14], [15]. The issues can be summarized as two primary areas. On the one hand, as the depth of convolutional neural networks increases, the detailed information captured in shallow layers gradually diminishes during layer-by-layer propagation. This phenomenon is particularly acute in underwater environments. Underwater images are often degraded by light attenuation, color distortion, and scattering, resulting in inherently blurred object edges and textures. Consequently, these already weak shallow features are further attenuated during propagation. The deep features are semantically strong, but they struggle to preserve the spatial locations and fine-grained structures of the objects. In contrast, while the loss of shallow-level details in land-based scenes may have a limited impact owing to ample data and accurate annotations. The absence of this information in low-visibility underwater settings exponentially increases detection difficulty, thereby significantly reducing identification and localization accuracy. On the other hand, recent improvements, such as RT-DETR [11] and DETR with Improved deNoising anchorboxes (DINO) [16], have optimized the DETR framework have focused on real-time performance and multimodal fusion, thereby enhancing detection efficiency and accuracy to some certain extent. However, these improvements primarily augment multimodal information and fail to fundamentally resolve the critical issues of shallow feature attenuation and insufficient fine-grained foreground-background discrimination in underwater imagery.

D. Problem Definition

Comprehensively analyzing the aforementioned points, we identified two core technical challenges that require focused attention in UOD. First, as network depth increases, shallow

feature information gradually degrades during transmission, leading to a loss of critical details. Although deep features provide rich semantic information, lack a detailed characterization of spatial location information. Second, both shallow and deep features play equally important roles in underwater scenes. When this shallow information is lost or ineffectively integrated with deep features, the model's ability to distinguish foreground objects from complex backgrounds is significantly diminished. Therefore, we address the key technical challenge of enabling guided learning across shallow and deep features to enhance the fine-grained distinction between the foreground and background.

E. Proposed Solutions

In this article, we propose the DETR with foreground-guided feature refinement network (DFFR). Inspired by recent advances in detection frameworks, some studies focus on real-time performance and multimodal fusion, while others adopt deeper convolutional architectures to enhance feature representation [11], [14], [15], [16]. In contrast, our work explicitly models the guidance relationship between shallow and deep features. Meanwhile, building upon the DETR architecture, we introduce a foreground-guided feature refinement (FFR) mechanism. Specifically, we first employ the lightweight and efficient high performance GPU network V2 (HGNetV2) as the backbone to comprehensively extract multiscale deep semantic features from images, thereby supplying abundant semantic information for downstream detection tasks. Building on the features captured from the backbone, we propose the FFR module to enhance foreground representation. Subsequently, the location perception fusion (LPF) strategy effectively integrates multiscale semantic features to improve feature consistency and robustness to scale changes. Finally, we introduce the fine-grained distribution refinement (FDR) [17] module within the transformer decoder to iteratively refine the feature distribution via a residual approach, yielding a finer intermediate representation.

The key contributions presented in this work are as follows.

- 1) We designed a DFFR network for UOD. It strengthens guided interactions between shallow and deep features to enhance fine-grained foreground-background discrimination, achieving high accuracy in both underwater and land scenarios.
- 2) We proposed a FFR module to explicitly model interactions between shallow and deep features via high-level contextual guidance. This module enhances foreground representation by improving foreground-background discrimination in complex underwater environments.
- 3) We designed a location-aware feature refinement strategy that preserves original feature distributions by avoiding compressive operations, such as global pooling. This strategy directly models higher order semantic dependencies and spatial contextual relationships, enabling fine-grained and comprehensive feature interactions.

The rest of this article is organized as follows. Section II reviews relevant research on object detection. Section III describes the proposed DFFR architecture. Section IV reports and

discusses the experimental results. Finally, Section V concludes this article and suggests directions for future work.

II. RELATED WORKS

This section systematically introduces related works covering two-stage and one-stage detectors, in addition to DETR and its transformer-based improvements. The subsequent paragraphs detail these methods.

A. Two-Stage and One-Stage Detectors

Early convolution-based object detection frameworks are conventionally classified into one-stage and two-stage paradigms, both of which predominantly depend on manually designed anchors or reference points. Representative two-stage methods, such as Faster Regions with Convolutional Neural Networks (R-CNN) [18], first generate a set of candidate regions and then perform classification and bounding-box regression for each area. Although these methods achieve high detection accuracy, their complex architectures and multistage processes limit real-time applicability. By contrast, single-stage detectors like YOLO [4], [5], [6] directly perform classification and bounding box regression on dense image locations, offering simpler architecture, higher efficiency, and faster detection. Nevertheless, YOLO and related single-stage frameworks are fundamentally built upon convolutional neural networks, which primarily focus on local receptive fields. This intrinsic locality constrains their ability to capture long-range dependencies and global contextual cues. Such limitations are particularly detrimental in underwater scenarios. Consequently, CNN-based YOLO detectors often underperform in detecting small or ambiguous objects and suffer from feature degradation in visually complex underwater environments. To alleviate these limitations, a variety of enhancement strategies have been investigated, including the use of deeper backbone networks, multiscale feature aggregation schemes, and the incorporation of attention mechanisms [15], [19], [20]. These approaches collectively aim to enrich feature representations and thereby improve detection performance. While such strategies can partially address challenges related to local feature extraction and scale variation, these approaches remain inherently limited in modeling holistic contextual information and capturing long-range dependencies.

In comparison, DETRs inherently model global interactions, allowing each spatial location to consider information from all other positions [8]. This capability enables effective capture of long-range dependencies and contextual relationships. Given the limited ability of existing methods to capture semantic and contextual information in complex environments, recent research has increasingly explored transformer-based object detection frameworks. These frameworks leverage their inherent global modeling capacity and feature refinement capabilities to overcome this bottleneck.

B. DETR and Its Variants

As a strong competitor to the YOLO series, DETR leverages the powerful global modeling capabilities of the transformer

architecture to achieve direct set predictions. This approach obviates the need for heuristic components, such as manually designed anchors and NMS, thereby streamlining the detection pipeline. However, the original DETR suffers from slow convergence during training and unclear query semantics, limiting its practical generalization [8], [21]. Subsequent research has tackled these issues primarily by investigating techniques to accelerate convergence [9], [10], [22], [23], [24]. Some existing approaches aim to alleviate the inefficiency of the decoder cross-attention. For example, Sun et al. [23] proposed a DETR model that consists solely of an encoder, omitting the decoder entirely. Dai et al. [25] introduced a dynamic decoder designed to concentrate on salient regions across multiple feature scales. Another category of studies aims to provide deeper insights into the function of decoder queries within DETR. Dynamic anchor box (DAB)-DETR [10] reformulates the DETR query as a DAB, effectively bridging the gap between traditional anchor-box-based detectors and DETR-style models. Building upon this, DN-DETR [9] further addresses the instability of Bipartite Graph Matching by employing denoising techniques. Moreover, DINO [16] integrates DN-DETR [9], DAB-DETR [10], and the deformable attention mechanism to reduce the model parameters while maintaining competitive performance. In addition, RT-DETR [11] achieves a good balance for the DETR framework in real-time detection scenarios through a lightweight backbone network and an efficient decoder structure design.

However, applying these mainstream DETR variants to underwater scenes reveals significant limitations. For instance, DINO [16] possesses multiscale fusion and global-local modeling capabilities. Under low-contrast noise interference, it easily leads to blurred object edges. DN-DETR [9] employs a denoising mechanism to enhance training stability and convergence speed, but it lacks targeted solutions for the blurred edges and uneven illumination commonly found in underwater objects. RT-DETR [11] excels in real-time performance, achieving fast inference speeds through a lightweight backbone and efficient decoder architecture. Similarly, its accuracy remains limited when detecting small objects in complex backgrounds. By contrast, our proposed DFFR introduces a FFR and LPF mechanism, significantly enhancing the ability to distinguish underwater objects against complex underwater backgrounds while maintaining accuracy and efficiency.

III. PROPOSED METHOD

DFFR is an end-to-end underwater object detector, which is designed to address background interference and the inherently low-texture characteristics of underwater objects. In this section, we present a detailed description of the main components of DFFR.

A. Overview

Considering the pronounced background interference and the intrinsically low-texture nature of underwater objects, distinguishing foreground objects from cluttered backgrounds poses a considerable challenge. This shortcoming considerably impacts the precision and robustness of UOD systems.

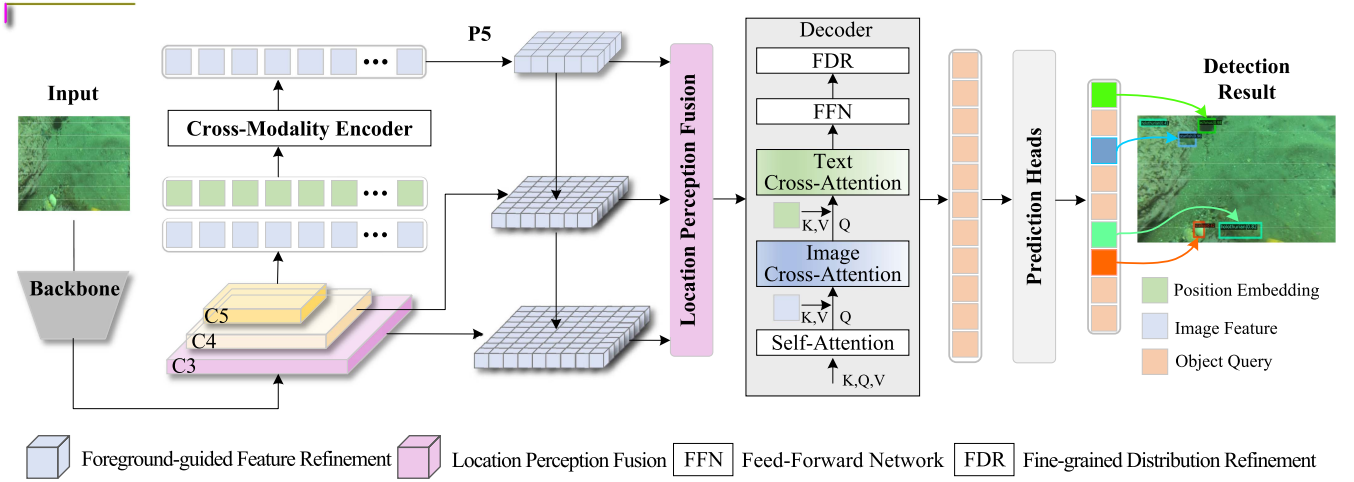


Fig. 2. Overall architecture of the proposed DFFR. The framework comprises the HGNetV2 backbone, a cross-modality encoder, a multipath feature fusion module, and a decoder equipped with auxiliary branches. The process begins by extracting multiscale features (C3, C4, and C5) from the HGNetV2 backbone and unify them into a high-level semantic representation (P5) via the encoder. Subsequently, the FFR module and LPF strategy perform semantic refinement to generate a more discriminative feature map. Finally, the decoder incorporates a self-attention mechanism, graph-based cross-attention, text-based cross-attention, a feed-forward network, and the FDR module to capture global dependencies and improve regression accuracy. Through the collaboration of these modules, DFFR produces robust and comprehensive detection results.

To mitigate these influences, we propose DFFR. The overall methodological framework is illustrated in Fig. 2. Specifically, we adopt HGNetV2 as the backbone network, owing to its compact architecture and computational efficiency. This backbone achieves reduced model complexity without sacrificing critical image structural information. Subsequently, we input the multiscale features extracted from stages C3, C4, and C5 of the backbone network into the cross-modality encoder [16], which uniformly converts them into the high-level semantic feature map P5. On this basis, we adopt a top-down and parallel fusion path structure to integrate high-level semantic and underlying detail information.

In particular, the FFR module is performed on the object region (*cf.*, Section III-B). Following this, the LPF strategy performs fine-grained multiscale feature modeling to produce a more compact and highly discriminative feature map (*cf.*, Section III-C). Finally, the refined multiscale features are fed into a decoder with auxiliary prediction branches to achieve fine modeling and prediction of the object through a multilayer decoding process. During the overall network training process, multitask loss function co-optimization (*cf.*, Section III-D) is employed to enhance the performance and robustness of UOD significantly.

B. Foreground-Guided Feature Refinement

To effectively enhance the feature representation capability in the foreground region and achieve more accurate object recognition, we propose the FFR module. As shown in Fig. 3, the FFR module has two input branches. One branch processes the C3–C5 features through a cross-modality encoder [16] to generate P5, serving as a high-level semantic feature. Another set of raw feature maps from the current layer, C3 and C4. First, we perform a scale transformation on the high-level features

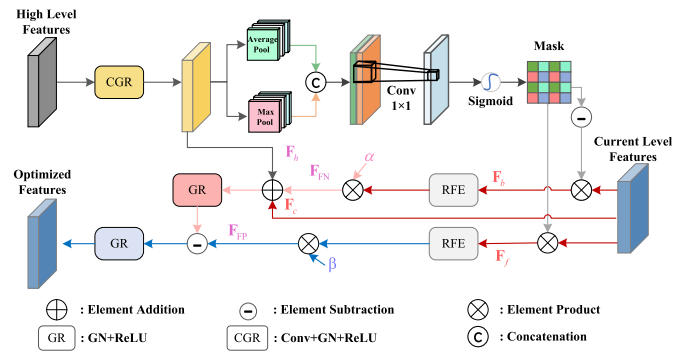


Fig. 3. Schematic illustration of the FFR module architecture. The module integrates features from the current layer with high level features information and generates spatial attention maps to emphasize foreground regions, thereby guiding the network to focus on salient foreground areas.

to align their spatial resolution with that of the current layer, enabling effective cross-level feature interaction. The aligned high-level features are then passed through max pooling and average pooling to generate spatial attention cues, which are further refined through a convolutional operation. Formally, given the aligned high-level feature $\mathbf{F}$, the attention generation process is defined as

$$\text{Mask} = \sigma(\text{Conv}(\text{MaxPool}(\mathbf{F}) + \text{AveragePool}(\mathbf{F}))) \quad (1)$$

where $\mathbf{F}$ denotes the aligned high-level feature map, MaxPool and AveragePool represent max pooling and average pooling operations, respectively. σ is the sigmoid function used to normalize the fused response into an attention mask. Subsequently, the generated attention map is element-wise multiplied with the current-layer feature map to highlight the responses in the foreground regions. Meanwhile, its complementary form, is multiplied with the same feature map to extract background

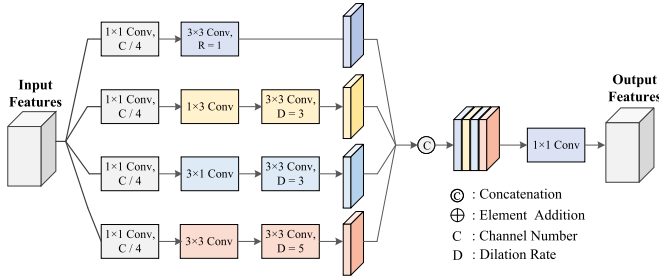


Fig. 4. Overall architecture of the RFE module. The module consists of four parallel branches combining inception and dilated convolution to enlarge the receptive field and capture multiscale context.

information. Through this complementary attention decomposition, the current-layer features are naturally separated into foreground features $\mathbf{F}_f$ and background features $\mathbf{F}_b$.

To further enhance the discrimination between $\mathbf{F}_f$ and $\mathbf{F}_b$, we incorporate a receptive field enhancement (RFE) module. As illustrated in Fig. 4, RFE is inspired by the inception architecture [26], and its design exhibits two key characteristics. First, it adopts a multibranch structure composed of convolutional layers with different kernel sizes, enabling the module to capture visual patterns at multiple spatial scales. This heterogeneous branching design is consistent with the philosophy of the Inception family and effectively enriches the diversity of receptive fields. Second, RFE integrates dilated convolutions with varying dilation rates, a technique previously proven effective in segmentation frameworks, such as DeepLab. The use of dilated convolutions allows the receptive field to expand without a proportional increase in computational cost, and provides a complementary effect similar to that of deformable convolutions by enabling the network to model long-range spatial dependencies. Within RFE, each branch is assigned a different dilation rate, allowing the module to selectively encode contextual cues from near, mid-range, and distant spatial regions. The outputs from all branches, each defined by distinct kernel sizes and dilation rates, are concatenated and subsequently integrated to produce a unified feature representation. This hierarchical fusion markedly enhances the capacity of the module to capture multiscale semantic structures and strengthens the contextual differentiation between foreground features and background features. Consequently, RFE provides richer structural information and strengthens the expressiveness of the refined features, ultimately contributing to more robust downstream object recognition. Following the enhancement of $\mathbf{F}_f$ and $\mathbf{F}_b$, we further apply residual augmentation to $\mathbf{F}_b$ by incorporating high-level bootstrap semantics, thereby improving contextual understanding. In addition, we introduce learnable weights α and β to adaptively regulate the spatial fusion strength between foreground and background features. The overall fusion process is expressed as

$$\mathbf{F}_r = \text{GR}(\text{GR}(\mathbf{F}_c + \mathbf{F}_h + \alpha\mathbf{F}_{\text{FN}}) - \beta\mathbf{F}_{\text{FP}}) \quad (2)$$

where $\mathbf{F}_r$ denotes the reconstructed feature map guided by the refined foreground. $\mathbf{F}_c$ and $\mathbf{F}_h$ denote the features from the current layer and the higher layer, respectively. α and β are learnable scaling parameters initialized to 1, which control the

contributions of the false-negative and false-positive components, respectively. $\mathbf{F}_{\text{FN}}$ and $\mathbf{F}_{\text{FP}}$ correspond to the false-negative and false-positive components. The former captures foreground regions with insufficient activation, while the latter represents background regions erroneously highlighted as foreground. Both components are obtained using a supervision-driven response mask derived from ground-truth annotations, enabling $\mathbf{F}_r$ to be more responsive to true foreground and more robust against background noise. GR denotes the combination of group normalization and rectified linear unit (ReLU) activation.

Ultimately, integrating the FFR and RFE modules equips the network with improved selectivity along salient dimensions, while simultaneously enhancing spatial perception and contextual representation. This joint refinement facilitates the generation of more compact and discriminative feature representations.

C. Location Perception Fusion

In real underwater environments, relying solely on fused features is inadequate to resolve the ambiguity and uncertainty inherent in underwater objects. This limitation often results in inaccurate localization. Inspired by Fu et al. [27], we propose the LPF strategy, which actively synchronizes the discriminative capability and localization sensitivity of feature maps. The LPF employs a dual-path fine-grained modeling mechanism that integrates channel dependency encoding (CDE) and contextual spatial encoding. By jointly capturing semantic dependencies and spatial structural cues, the LPF enhances both the discriminative capacity and the localization sensitivity of the feature representation.

The CDE unit is designed to capture higher order dependencies across channels, thereby enhancing the discriminability of semantic feature representations. Given the output feature $\mathbf{F}_r \in \mathbb{R}^{C \times H \times W}$ from the FFR module, $\mathbf{F}_r$ is first processed by shared convolutional transformations and then reshaped into a matrix of size $\mathbb{R}^{C \times N}$, where $N = H \times W$ denotes the number of spatial positions.

To model long-range spatial correlation, we compute a spatial attention map $\mathbf{S} \in \mathbb{R}^{N \times N}$ using a softmax normalization. Let $\mathbf{b}_i$ and $\mathbf{c}_j$ denote the projected feature embeddings at the i th and j th spatial positions, obtained through two learnable linear projections. The attention weight between positions i and j is formulated as

$$\mathbf{S}_{ji} = \frac{\exp(\mathbf{b}_i^\top \cdot \mathbf{c}_j)}{\sum_{k=1}^N \exp(\mathbf{b}_k^\top \cdot \mathbf{c}_j)} \quad (3)$$

where $\mathbf{b}_i^\top \cdot \mathbf{c}_j$ measures the affinity between two spatial locations, and the normalized attention score $\mathbf{S}_{ji}$ reflects their spatial dependency strength. This attention map is subsequently used to reweight the feature responses for enhanced spatial representation learning.

Meanwhile, $\mathbf{F}_r$ is transformed by a third convolution to obtain $\mathbf{D} \in \mathbb{R}^{C \times H \times W}$, which is similarly reshaped into $\mathbb{R}^{C \times N}$ and then multiplied by the transpose of the attention map $\mathbf{S}$. It is then added element-wise to the original feature $\mathbf{F}_r$. Finally, by introducing a multihead channel interaction unit to further model

global interchannel dependencies, the output spatially enhanced feature $\mathbf{F}_{\text{CDE}}$ is obtained, which is mathematically expressed as

$$\mathbf{F}_{\text{CDE}} = \text{MHCI} \left(\sum_{i=1}^N (\mathbf{S}_{ji} \cdot \mathbf{D}_i) + \mathbf{F}_r \right) \quad (4)$$

where MHCI denotes the partitioning of channels into multiple subspace, each independently computing attention and subsequently aggregating the results.

In contrast to the CDE module, we directly compute the channel attention map $\mathbf{M} \in \mathbb{R}^{C \times C}$ from $\mathbf{F}_r$. Specifically, we directly pass $\mathbf{F}_r \in \mathbb{R}^{C \times H \times W}$ through reshape to $\mathbb{R}^{C \times N}$, and then we perform matrix multiplication on $\mathbf{F}_r$ and its transpose. Channel attention mapping $\mathbf{M} \in \mathbb{R}^{C \times C}$ is obtained by softmax layer, which is expressed as

$$\mathbf{M}_{ji} = \frac{\exp(\mathbf{F}_{ri} \cdot \mathbf{F}_{rj})}{\sum_{i=1}^C \exp(\mathbf{F}_{ri} \cdot \mathbf{F}_{rj})} \quad (5)$$

where $\mathbf{M}_{ji}$ denotes the effect of the i th channel on the j th channel. We then perform a matrix multiplication operation on the transpose of the attention map $\mathbf{M}$ with the feature map $\mathbf{F}_r$ and reshape the result to $\mathbb{R}^{C \times H \times W}$. Finally, we perform an element-by-element summation operation with the original features to obtain the final output feature $\mathbf{F}_{\text{CSE}} \in \mathbb{R}^{C \times H \times W}$, which is expressed as

$$\mathbf{F}_{\text{CSE}} = \text{MHCI} \left(\sum_{i=1}^N (\mathbf{M}_{ji} \cdot \mathbf{F}_{ri}) + \mathbf{F}_{rj} \right). \quad (6)$$

The above formula shows that the final feature of each channel is the sum of all channel features and its original feature, thus establishing an upper-distance semantic dependency between feature maps and improving the discriminative nature of the features.

Notably, we model interchannel relationships without applying compressive operations, such as global pooling, convolutional mapping, or dimensionality reduction. We directly establish response correlations between channels within the original feature space, thereby maximizing the preservation of the feature integrity and distributional information. This design effectively mitigates spatial information loss from feature compression and enables more comprehensive, fine-grained channel dependency modeling.

D. Loss Function

To jointly optimize classification and regression tasks, we adopt the multinomial joint loss strategy [11]. This strategy provides fine-grained supervision for the prediction results and guides the model to progressively refine boundary distributions and semantic representations across multiple layers. In order to further enhance the learning quality of deep features and maintain multiscale semantic consistency, we introduce supervised signals at all layers of the decoder and its auxiliary branches. Consequently, this multiobjective loss framework shows superior generalization and robustness in underwater complex background environments, providing a strong optimization basis for

subsequent detection tasks. The total loss function is formulated as

$$\mathcal{L}_{\text{TOTAL}} = \mathcal{L}_{\text{CLS}} + \mathcal{L}_{\text{REG}} + \lambda_1 \mathcal{L}_{\text{FGL}} + \lambda_2 \mathcal{L}_{\text{DDF}} \quad (7)$$

where $\mathcal{L}_{\text{CLS}}$ denotes the classification loss, $\mathcal{L}_{\text{REG}}$ is the bounding box regression loss, $\mathcal{L}_{\text{FGL}}$ denotes the fine-grained localization loss, $\mathcal{L}_{\text{DDF}}$ is the decoupled distillation focal loss, and λ_1 and λ_2 are the weighting coefficients of the individual branching loss terms. Following the general settings reported in [28], we set $\lambda_1 = 1$ and $\lambda_2 = 2$ in all experiments to ensure reproducibility and consistency.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we conduct extensive experiments to validate the effectiveness and robustness of our proposed DFFR. We evaluate its performance on three representative benchmark datasets, including DUO [29], UTDAC [30], and COCO [31]. Detailed descriptions of the datasets, experimental settings, and evaluation metrics are provided. In addition, we perform ablation studies on key components to investigate their impact on the overall detection performance.

A. Datasets

DUO dataset: DUO [29] is a widely adopted benchmark dataset developed for UOD. It comprises 7782 images, which are divided into 6671 for training and 1111 for testing, with roughly 10% of the training images randomly chosen as a validation set. The dataset features diverse underwater environments and provides annotations for 74 515 objects of holothurians, echinus, starfish, and scallops.

UTDAC dataset: UTDAC [30] is an underwater dataset released as part of the Underwater Detection Algorithm Competition 2020. The dataset contains 5168 training and 1293 validation images, annotated for echinus, holothurian, starfish, and scallop. It provides images at four resolutions (3840×2160 , 1920×1080 , 720×405 , and 586×480), reflecting the challenges of varying image quality in real underwater environments.

COCO dataset: COCO [31] is a widely used benchmark for general object detection and segmentation. It consists of 118 287 training images and 5000 test images, covering 80 object categories including humans, animals, and vehicles. This dataset enables us to validate the performance and generalization ability of our proposed method on conventional vision tasks.

B. Experimental Details and Evaluation Metrics

Experimental details: To obtain the main experimental results, we adopt the HGNetV2 backbone. The stem and normalization layers are frozen during training. All input images are resized to 640×640 , and a multiscale training scheme is used with a base resolution of 640 and three repeated samples. Data augmentation includes random horizontal flipping with a probability of 0.5, random cropping to 384×384 without empty regions, and random brightness and contrast adjustments with a limit of 0.2 and a probability of 0.5. The training uses a batch size of 32, while validation uses a batch size of 64. AdamW is employed as

TABLE I
COMPARISON OF OUR PROPOSED DFFR WITH OTHER STATE OF THE ART (SOTA) METHODS ON BENCHMARK DUO [29] AND UTDAC [30] DATASETS

Methods	DUO						UTDAC						Params(M)	GFLOPs	FPS
	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L			
Faster-RCNN [18]	61.3	81.8	69.2	46.2	62.8	60.3	44.5	80.9	44.1	20.0	39.0	50.8	206.68	41.14	30.7
FCOS [28]	61.5	82.2	68.2	49.7	63.5	59.9	43.8	81.1	43.0	19.9	38.2	50.4	196.81	31.84	31.2
Free-Anchor [32]	62.2	81.2	70.8	45.1	64.1	60.0	46.3	82.3	46.9	21.0	40.5	52.6	205.69	36.17	24.3
Sparse-RCNN [24]	64.3	81.9	71.5	41.0	66.1	63.1	37.4	70.4	35.9	17.7	33.3	43.0	149.90	105.95	31.6
TOOD [19]	66.2	84.0	73.2	55.0	68.4	64.2	39.4	75.2	38.4	20.5	35.6	45.8	180.66	31.80	25.5
B-RCNN [12]	66.1	82.9	73.7	50.1	68.3	64.4	48.5	82.4	52.5	21.1	41.1	55.0	173.20	45.91	30.0
DETR [8]	61.5	81.5	70.9	49.5	66.5	62.2	46.6	84.1	47.0	24.1	42.4	51.9	41.00	86.00	42.4
YOLO v10 [4]	62.3	81.6	69.1	51.9	60.8	60.9	49.2	82.3	52.0	22.5	42.3	54.2	2.71	8.40	110.6
YOLO 11 [5]	63.1	82.7	70.7	52.1	65.1	61.4	49.9	83.5	52.7	22.7	42.9	54.9	2.59	6.40	125.3
YOLO v12 [6]	65.2	83.7	72.2	52.6	66.6	64.5	50.8	84.4	54.2	23.5	43.7	55.8	2.57	6.50	122.7
RFTM [33]	60.4	80.3	69.1	38.7	60.8	60.9	47.4	80.3	47.6	20.3	40.4	54.1	234.47	75.59	18.9
DINO [16]	66.4	85.2	73.5	52.7	66.8	65.2	40.4	76.4	39.3	21.6	36.8	46.7	47.29	128.64	49.6
UDMdet [15]	67.2	84.6	74.1	50.1	68.3	64.4	45.0	80.6	46.2	18.3	40.3	50.9	234.47	70.10	23.3
CIDNet [14]	68.8	86.5	75.5	56.5	70.6	67.3	49.4	85.1	52.5	24.3	44.4	55.2	324.48	62.91	42.3
RT-DETR [11]	69.1	86.2	75.7	54.8	70.1	68.8	50.7	84.5	55.8	23.7	45.9	56.6	42.00	152.63	48.8
DFFR (Ours)	70.9	88.9	77.4	56.5	71.3	69.4	52.3	86.2	57.4	25.6	47.8	58.1	31.82	100.17	53.6

All the bold values mean the best result.

the optimizer with a grouped learning-rate strategy. The learning rate for nonnormalized backbone layers is set to 1.25×10^{-5} , and the base learning rate is 2.5×10^{-4} . Weight decay is fixed at 1.25×10^{-4} , and the momentum coefficients β are set to [0.9, 0.999]. The model is trained for 80 epochs, with the final 8 epochs used for fine-tuning, during which the learning rate is reduced by a factor of 10. An Exponential Moving Average is applied to stabilize parameters and accelerate convergence. Our DFFR and the compared approaches were executed on a PC equipped with an Intel Core i7-12700KF CPU, 32GB of system memory, and an NVIDIA GeForce 4060 Ti GPU with 16GB of video memory.

Evaluation metrics: We evaluate DFFR using COCO's standard metrics. In addition, we report efficiency indicators, including model parameter counts, computational complexity measured in giga floating point operations per second (GFLOPs), and inference speed measured in frames per second (FPS). Collectively, these metrics enable a rigorous assessment of the computational requirements, efficiency, and real-time capabilities of the proposed DFFR.

C. Comprehensive Evaluation on Underwater Dataset

Quantitative evaluation: We performed comparative experiments with several mainstream detection frameworks on two representative underwater datasets, DUO [29] and UTDAC [30]. The corresponding results are presented in Table I. On the DUO [29] dataset, DFFR attains an average precision (AP) of 70.9%. It markedly surpasses several leading methods, including CIDNet [14] with an AP of 68.8%, UDMdet [15] with an AP of 67.2%, and TOOD [19] with an AP of 66.2%. In particular, DFFR achieves 56.5% in AP_S for small object detection. This represents a significant improvement over both Faster R-CNN [18] with an AP_S of 46.2% and Free-Anchor [32] with an AP_S of 45.1%. Such improvement can be attributed to the proposed FFR module, which exploits high-level

contextual information to strengthen the representation of foreground objects. DFFR also demonstrates strong performance on the UTDAC [30] dataset. In the context of more complex environments, the proposed method shows good robustness, mainly attributed to the introduction of the cross-modality encoder. Meanwhile, the LPF strategy further enhances spatial perception across multiscale features and mitigates localization bias at varying scales.

Apart from achieving high detection accuracy, DFFR exhibits strong computational efficiency and real-time inference capability, which are essential for practical deployment. As indicated in Table I, the model contains 31.82 M parameters and 100.17 GFLOPs, and attains 53.6 FPS, falling within the range for industrial applications, including AUVs and other resource-constrained systems. This balanced tradeoff demonstrates the suitability of DFFR for practical scenarios.

Qualitative evaluation: In UOD quality assessment, detection accuracy is always the primary criterion for measuring the performance of various methods. To this end, we selected representative images from the DUO [29] and UTDAC [30] datasets, covering three typical interference scenarios: blurring, low visibility, and occlusion. As illustrated in Fig. 5, the results of the qualitative comparison of CIDNet with other SOTA methods. For example, TOOD [19], B-RCNN [12], CIDNet [14], and UDMdet [15] exhibit varying degrees of misdetection. Moreover, despite being a state-of-the-art general-purpose detector, YOLO v12 [6] continues to exhibit both false negatives and false positives in underwater-specific scenarios. Nevertheless, our method achieved accurate detection in underwater scenarios.

D. Generalization Evaluation on the COCO Dataset

Quantitative evaluation: To assess generalization and cross-domain adaptability, we conducted evaluations on the standard COCO [31] dataset and compared DFFR with mainstream detectors. The corresponding quantitative results are reported in

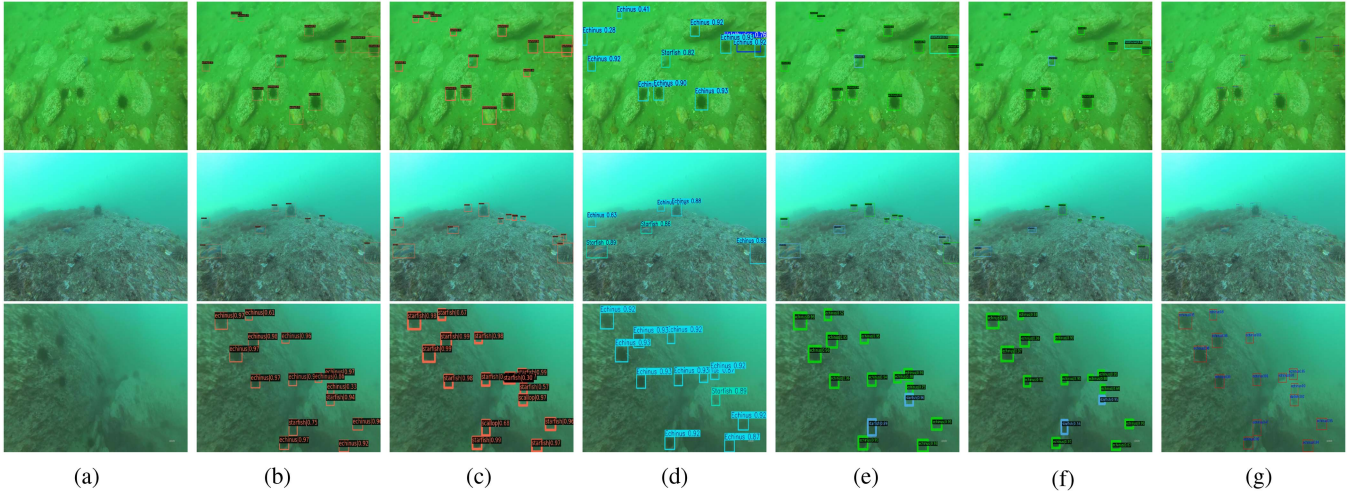


Fig. 5. Qualitative comparison results on the underwater benchmark DUO [29] and UTDAC [30] datasets. Our method demonstrates superior detection in terms of object integrity, localization accuracy, and boundary clarity. (a) Input. (b) TOOD [19]. (c) B-RCNN [12]. (d) YOLO v12 [6]. (e) UDMdet [15]. (f) CIDNet [14]. (g) DFFR.

TABLE II
COMPARISON OF OUR PROPOSED DFFR WITH OTHER SOTA METHODS ON THE COCO [31] DATASET

Methods	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Faster-RCNN [18]	37.4	58.1	40.4	21.2	41.0	48.1
FCOS [28]	38.5	57.7	41.0	21.9	42.8	48.6
Free-Anchor [32]	38.9	57.9	41.5	21.1	41.8	49.0
Sparse-RCNN [24]	39.9	58.6	42.7	21.1	41.5	53.6
TOOD [19]	42.0	59.5	45.5	23.8	44.6	53.2
DETR [8]	43.5	63.8	46.4	21.9	48.0	61.8
YOLO v10 [4]	38.0	52.1	40.6	12.5	35.0	53.5
YOLO 11 [5]	38.8	54.3	42.2	13.6	35.1	55.1
YOLO v12 [6]	40.2	55.4	43.5	14.0	36.3	56.4
RFTM [33]	37.5	58.1	40.2	21.1	40.2	47.1
DINO [16]	50.9	69.0	55.3	34.6	54.1	64.6
UDMdet [15]	42.6	60.3	46.2	24.5	45.6	53.9
CIDNet [14]	45.4	63.4	49.4	28.1	49.7	58.5
RT-DETR [11]	53.1	71.3	57.7	34.8	58.0	70.0
Ours	52.4	71.6	57.4	35.2	58.4	71.5

All the bold values mean the best result.

Table II. Despite being designed for underwater environments, DFFR attains strong performance with an AP of 52.4%. It outperforms classical two-stage detectors, such as Faster R-CNN [18], and also surpasses anchor-free methods including FCOS [28] and YOLO v12 [6]. Compared with recent DETRs, such as DINO [16] and RT-DETR [11], DFFR remains highly competitive. Notably, the proposed method achieves an AP₇₅ of 57.4%, which is on par with the 57.7% reported by RT-DETR [11]. These results underscored the ability of DFFR to deliver state-of-the-art performance.

Qualitative evaluation: To further evaluate the generalization capability of DFFR, we selected images from the COCO dataset that resemble underwater environments. The detection results of DFFR were compared with those of YOLO v10 [4], YOLO

TABLE III
ABLATION EXPERIMENTS OF OUR PROPOSED DFFR ON THE DUO [29] DATASET

Methods	FFR	RFE	LPF	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
HGNetV2-L	×	×	×	57.5	80.0	65.0	44.7	56.8	63.1
Group A	✓	×	×	70.0	88.6	77.1	54.6	70.2	69.0
Group B	×	✓	×	69.6	88.0	76.5	54.1	70.8	68.6
Group C	✓	✓	×	70.5	88.8	77.2	54.2	71.1	68.9
Group D	×	×	✓	65.4	84.6	72.4	52.4	66.8	64.6
Group E	✓	✓	✓	70.9	88.9	77.4	56.5	71.3	69.4

All the bold values mean the best result.

11 [5], YOLO v12 [6], DETR [8], and CIDNet [14], as shown in Fig. 6. In these scenarios, DFFR achieves more accurate multi-scale object recognition, clearer boundary delineation, and significantly reduced false detections, while demonstrating strong robustness for small and occluded objects. These qualitative results highlight that DFFR not only excels in standard evaluation settings, but also possesses strong practical applicability. Consequently, DFFR is suitable for deployment in real-world underwater and cross-domain detection tasks.

E. Ablation Study

To comprehensively evaluate the effectiveness of the proposed approach, we adopt HGNetV2-L as the baseline model and conduct ablation experiments by progressively incorporating each module of the DFFR framework. All experiments are performed on the DUO [29] dataset, as summarized in Table III. We designed five groups (A-E) to isolate the effects of different module combinations. Group A includes only the FFR module, while the RFE module responsible for foreground and background enhancement is removed. Group B retains only the RFE module, which directly enhances the backbone features without involving the foreground-background refinement pathway. Group C

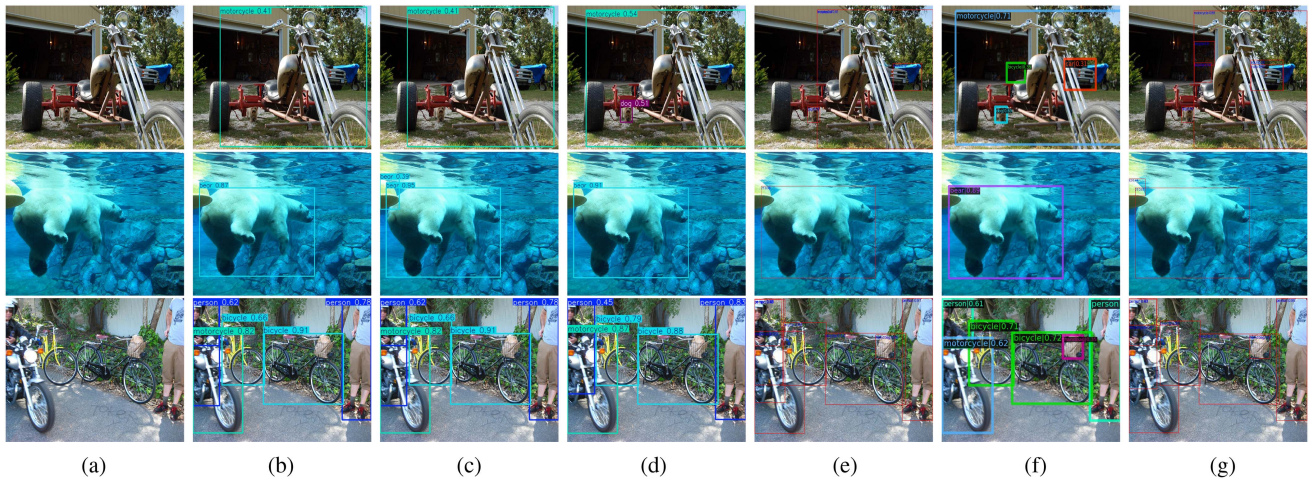


Fig. 6. Qualitative comparison of typical natural scene detection results on the publicly available COCO [31] dataset. Our method is effective in reducing the number of missed and false detections when dealing with small, similar, dense and occluded objects. (a) Input. (b) YOLO v10 [4]. (c) YOLO 11 [5]. (d) YOLO v12 [6]. (e) DETR [8]. (f) CIDNet [14]. (g) DFFR.

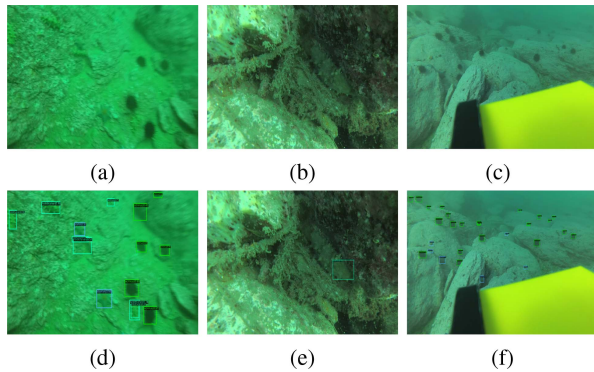


Fig. 7. Failure cases. The first row illustrates challenging scenarios: (a) Blurred region, (b) camouflaged object, and (c) small objects. The second row shows the corresponding detection results: (d) Detection result for (a), (e) detection result for (b), and (f) detection result for (c). These results highlight missed detections or low-confidence predictions under extreme underwater conditions.

integrates both FFR and RFE, allowing us to examine their complementary effects on feature refinement. Group D activates only the LPF strategy, enabling evaluation of its contribution to multiscale feature fusion. Group E incorporates all three modules and corresponds to the full DFFR configuration. As detailed below, the performance of each configuration is further analyzed to quantify the individual and combined contributions of the proposed modules.

- 1) *Effectiveness of FFR*: As shown by the comparison between Groups A and B, incorporating the FFR module alone increases the overall AP to 70.0%, which is a 0.4% point improvement over the baseline without FFR. Notably, FFR exhibits a notable advantage in small object detection, where AP_S increases to 54.6%. This demonstrates that FFR effectively strengthens foreground representations, thereby alleviating the confusion between small objects and background clutter.
- 2) *Effectiveness of RFE*: From the comparison between Groups A and B, it can be observed that using RFE

alone achieves an overall AP of 69.6%, which is comparable to FFR. However, RFE performs slightly better for medium-scale objects, achieving an AP_M of 70.8%. This indicates that RFE is beneficial for multiscale receptive field modeling and enhances the discrimination capability for medium-sized objects. Furthermore, when RFE is combined with FFR in Group C, the overall AP increases to 70.5%, indicating that the two modules function in a complementary manner.

- 3) *Effectiveness of LPF*: As indicated by the comparison between Group D and the other groups, introducing LPF alone reduces the overall AP to 65.4%, implying that relying solely on multiscale information result in insufficient discriminative features. Nevertheless, AP_S remains at 52.4%, showing that LPF enhances the holistic contour representation of objects to some extent. When LPF is combined with FFR and RFE in Group E, the overall AP reaches the highest value of 70.9%, and AP_S increases to 56.5%. These results confirm that LPF provides a complementary contribution and strengthens the synergistic effect of multimodal feature fusion.

Through the above experiments, we can further verify the functional contributions of each component within the DFFR framework. The FFR module strengthens foreground representations by emphasizing salient object regions and suppressing background interference, which is particularly beneficial for small-object detection, as reflected in the improved AP_S . The RFE module, by expanding the receptive field through multi-branch and dilated convolutions, enhances multiscale contextual modeling and improves discrimination for medium-sized and context-rich objects, thereby complementing the refinement effect of FFR. Although the LPF alone contributes limited overall discriminative power, it captures holistic contours and low-frequency information that naturally align with the detailed refinement of FFR and the contextual enhancement of RFE. When combined, these modules exhibit a synergistic effect, leading to notable performance gains. The full configuration in Group E yields the highest accuracy across all metrics,

demonstrating that the joint design of these modules substantially enhances detection performance in complex underwater scenarios. Overall, the experimental results confirm the effectiveness of the proposed DFFR and highlight the complementary contributions of its individual modules.

V. CONCLUSION

In this article, we propose DFFR to facilitate collaborative learning between shallow and deep features, explicitly modeling their interactions to enhance fine-grained foreground-background discrimination. By mitigating shallow feature attenuation and strengthening feature integration, DFFR improves detection accuracy for both visible and subtle underwater objects. Specifically, FFR module strengthens foreground representations, effectively highlighting salient object regions and mitigating confusion with background clutter, which is particularly beneficial for small-object detection. RFE module expands the receptive field through multibranch and dilated convolutions, enhancing multiscale contextual modeling and improving discrimination for medium-sized and context-rich objects. LPF strategy preserves holistic object contours, complementing the effects of FFR and RFE to enable more robust multiscale feature fusion. Extensive experiments on the DUO [29], UTDAC [30], and COCO [31] benchmarks show that DFFR achieves competitive AP, AP₅₀, and AP₇₅, demonstrating strong generalization ability. Furthermore, the ablation study further substantiates the individual and synergistic contributions of the proposed modules, validating the overall effectiveness of the method.

Limitations: While DFFR demonstrates strong performance under standard conditions, it still exhibits limitations under extreme underwater scenarios. Fig. 7 shows representative failure cases. On the one hand, under highly blurred or turbid conditions, although DFFR integrates FFR and LPF, it may still misidentify objects, leading to inaccurate detections or low-confidence scores. On the other hand, small or distant objects occupy very few pixels, producing weak feature activations in the backbone and transformer layers, which reduces bounding box regression accuracy. In addition, the model being trained on medium-sized labeled datasets. While sufficient for initial training, these datasets may lack the diversity required to capture rare object instances or extreme underwater conditions. To alleviate these issues, we attempted integrating deblurring modules to enhance feature quality under turbid conditions. However, while image clarity improved, feature representation strength decreased, yielding limited performance gains. Similarly, incorporating synthetically generated extreme samples into the training set provides limited robustness gains, as the domain discrepancy between synthetic and real-world data restricts its overall effectiveness. These observations highlight the inherent challenges of extreme scenarios that remain difficult to fully address with the current model.

Future work: Building on the limitations identified, future work will focus on developing strategies to better handle extreme underwater scenarios. This includes designing modules that can enhance feature representation under highly turbid or blurred conditions without degrading backbone and transformer activations. In addition, exploring more diverse and realistic data

augmentation strategies, or domain adaptation techniques, could help bridge the gap between synthetic and real-world samples. Beyond robustness, we aim to develop compact and memory-efficient architectures that sustain high detection performance while enabling real-time inference on embedded and underwater platforms.

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Gaoli Zhao received the M.S. degree in software engineering from the School of Computer Science, Sichuan University, Chengdu, China, in 2011.

He is currently an Associate Professor with the School of Computer Science and Technology, Henan Institute of Science and Technology, Xinxiang, China. He has authored or coauthored more than 30 papers. His current research interests include deep learning-based image enhancement and object detection in computer vision.



Kefei Zhang (Student Member, IEEE) received the B.S. degree in computer science and technology from the School of Computer Science and Technology, Zhengzhou Technology and Business University, Zhengzhou, China, in 2024. He is currently working toward the M.S. degree in agricultural engineering and information technology with the School of Computer Science and Technology, Henan Institute of Science and Technology, Xinxiang, China.



Yuheng Wu received the B.S. degree in internet of things engineering from the Henan University of Science and Technology, Xinxiang, China, in 2022, where he is currently working toward the M.S. degree in electronic information with the School of Computer Science and Technology.



Zheng Liang received the Ph.D. degree in computer applied technology from the School of Information Science and Technology, Dalian Maritime University, Dalian, China, in 2021.

He is currently a Lecturer with the School of Internet, Anhui University, Hefei, China. His current research interests include image restoration, computer vision, and deep learning.



Wenyi Zhao received the Ph.D. degree in control science and engineering from the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing, China, in 2023.

He is currently a Postdoctoral Researcher with the School of Artificial Intelligence, Beijing University of Posts Telecommunications. His research interests include computer vision, machine learning, deep learning, contrastive learning, and self-supervised learning.



Weidong Zhang (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the School of Information Science and Technology, Dalian Maritime University, Dalian, China, in 2022.

He is currently a Postdoctoral Fellow with the Postdoctoral Innovation Practice Base of Henan Institute of Science and Technology, Luoyang, China, the Postdoctoral Workstation of Henan Bainong Seed Industry Limited Company, Xinxiang, China, and the College of Electrical and Information Engineering of Zhengzhou University, Zhengzhou, China. He is also a Professor with the School of Information Engineering, Henan Institute of Science and Technology. He has authored or coauthored more than 100 journal/conference papers on IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING INFORMATION, AAAI, etc. His current research interests include image restoration, intelligent agriculture, and deep learning.